Symbolic Mechanism Discovery via Best-First Search in Mathematical Operation Space

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Abstract

We introduce a framework for automated discovery of interpretable predictive formulas from labeled scientific data using best-first search through a space of mathematical operations. Given a feature matrix and binary labels, the method searches over compositions of arithmetic, transcendental, and conditional operations to find the closed-form formula that maximizes F1 score. The search is organized in phases of increasing complexity—single features, pairwise combinations, unary transforms, and triples—with GPU-vectorized evaluation enabling thousands of candidate formulas per second.

The framework provides two outputs: (i) the best interpretable formula at each depth, and (ii) the *formula ceiling*—the maximum F1 achievable by any closed-form expression of bounded complexity. Comparing this ceiling to a black-box ensemble quantifies the *irreducible complexity gap*: how much predictive structure resists symbolic expression.

Applied to predicting periodic orbits in the gravitational three-body problem (15,000 configurations, 23 Hessian-derived features, GPU-integrated ground truth), the method discovers that periodicity is predicted by the norm $\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2}$ (F1 = 0.48). The ceiling is stable across all tested operations at depths 1–3. A gradient-boosted ensemble trained on a separate 51,000-configuration dataset reaches F1 = 0.86, yielding an irreducible complexity gap of 0.38—meaning that no formula found within the bounded search space achieves more than 56% of the ensemble's predictive power.

Code and data: `https://github.com/ahb-sjsu/astar-symbolic-discovery`.

1 Introduction

A central goal of computational science is to discover interpretable laws from data. This problem has been addressed by genetic symbolic regression [1, 4], neural equation discovery [2], and classical AI systems [3]. Genetic programming explores large formula spaces but provides no optimality guarantee at a given depth and often produces bloated expressions. Neural methods discover accurate but opaque representations. Classical systems require domain-specific search heuristics.

We propose a structured search through mathematical operation space using *best-first search* with F1 score as the ranking criterion. The search is organized in phases of increasing formula complexity: single features (depth 1), pairwise combinations with binary operations (depth 2), unary transforms of top pairwise formulas (depth 3), and triple feature combinations (depth 3). At each phase, all candidate formulas are evaluated exhaustively, guaranteeing that the best formula at each depth is found exactly—not approximately.

This phased exhaustive approach avoids the admissibility issues of heuristic A* search on formula spaces (where defining an admissible heuristic for future F1 improvement is nontrivial)

while preserving the key practical benefit: the best formula within each phase's structure is identified by enumeration.

Our framework provides two outputs that existing methods do not:

1. The **best interpretable formula** at each complexity level, found by exhaustive search within that level.

2. The **formula ceiling**—the maximum F1 achievable by any closed-form expression of bounded depth. Comparing this ceiling to a black-box model quantifies the *irreducible complexity gap*, telling the practitioner when further symbolic search is futile.

2 Method

2.1 Problem Formulation

Definition 1 (Symbolic Prediction Problem)**.** *Given a feature matrix $X \in \mathbb{R}^{N\times d}$ and binary labels $y \in \{0,1\}^N$, find a formula $f : \mathbb{R}^d \to \mathbb{R}$ and threshold $\tau \in \mathbb{R}$ that maximize $F_1(\hat{y}, y)$ where $\hat{y}_i = \mathbf{1}[f(X_i) > \tau]$, subject to* $\mathrm{depth}(f) \leq D$.

2.2 Operation Registry

The search space is defined by three operation classes:

Leaf nodes (depth 0): Each of the d features x_j.

Unary operations (depth +1): $\{\log|\cdot|, \sqrt{|\cdot|}, (\cdot)^2, 1/(\cdot), -(\cdot), |\cdot|, \sigma(\cdot), \tanh(\cdot)\}$ (8 operations).

Binary operations (depth +1): $+$, $-$, $\times$, $\div$, max, min, hypot, $(\cdot - \cdot)^2$, harmonic mean, geometric mean (10 operations).

2.3 Phased Exhaustive Search

Rather than a single tree search, we organize the exploration into four phases (Algorithm 1). Each phase exhaustively evaluates all formulas at its complexity level.

Threshold sweep. For each candidate formula f, we compute $v_i = f(X_i)$ for all N samples, then evaluate F1 at 11 percentile thresholds $\{20, 30, 40, \ldots, 95\}$ of the value distribution. Both $v > \tau$ and $v < \tau$ predictors are tested. The best threshold is retained. This discrete sweep approximates but does not guarantee the globally optimal threshold; in practice, F1 is piecewise-constant in τ (changing only when τ crosses a data point), and our 11 percentiles sample this step function densely enough that the best F1 is found exactly for all formulas tested in our experiments (verified by exhaustive threshold search on a subset). The sweep is vectorized and adds negligible cost.

GPU acceleration. Features are stored as GPU arrays (CuPy). All formula evaluations and threshold sweeps are vectorized GPU operations. Phase 2 evaluates $d(d-1) \times |\mathcal{B}|$ formulas (e.g., $23 \times 22 \times 10 = 5{,}060$) in seconds.

Completeness within each phase. Phases 1–2 are exhaustive over all feature (pairs) and operations. Phase 3 applies all unary transforms to the top-K results from Phase 2. Phase 4 tests multiple combination patterns over the top-K single features. The search is complete at each phase for the specified formula structure; no formulas of that structure are missed.

2.4 Scalability

The formula count per phase is: Phase 1: $d(1+|\mathcal{U}|)$; Phase 2: $d(d-1)|\mathcal{B}|$; Phase 3: $K|\mathcal{U}|$; Phase 4: $\binom{K'}{3}P$. Here K is the number of top pairwise results carried forward, K' the top single features, and P the combination patterns.

Algorithm 1 Phased Exhaustive Symbolic Search

Require: Features $X \in \mathbb{R}^{N \times d}$, labels y, operations $\mathcal{U}$ (unary), $\mathcal{B}$ (binary)
Ensure: Best formula and F1 at each depth

Phase 1: Single features (depth 1)
for $j = 1, \ldots, d$ **do**
  Evaluate $F_1^*(x_j)$ via threshold sweep
$R_1 \leftarrow$ top-K features by F1

Phase 2: Pairwise combinations (depth 2)
for $x_i, x_j \in \{1, \ldots, d\}$, $b \in \mathcal{B}$ **do**
  Evaluate $F_1^*(b(x_i, x_j))$ via threshold sweep
$R_2 \leftarrow$ top-K pairwise formulas

Phase 3: Unary transforms (depth 3)
for $f \in R_2$, $u \in \mathcal{U}$ **do**
  Evaluate $F_1^*(u(f))$ via threshold sweep
$R_3 \leftarrow$ top-K depth-3 formulas

Phase 4: Triple combinations (depth 3)
for $x_i, x_j, x_k \in R_1$, pattern p **do**
  Evaluate $F_1^*(p(x_i, x_j, x_k))$ via threshold sweep
return best formula across all phases

For $d = 23$, $|\mathcal{B}| = 7$, $|\mathcal{U}| = 5$, $K = 30$, $K' = 8$, $P = 4$, the total is approximately 5,600 formulas—evaluated in under one minute on a single GPU. Composing Phase 2 with itself (all pairwise-of-pairwise formulas) would require $O(d^2(d-1)^2 \cdot |\mathcal{B}|^2) \approx 12$ million formulas, tractable with GPU vectorization but requiring hours. We report results through Phase 3 (depth 3); Phase 4 triples are also depth 3 but with a different structural template.

2.5 Formula Ceiling and Irreducible Complexity Gap

Definition 2 (Formula Ceiling)**.** *The formula ceiling C_D at depth D is the maximum F1 achieved by any formula of depth $\leq D$ found across all phases: $C_D = \max_{phases} F_1^*$.*

Definition 3 (Irreducible Complexity Gap)**.** *Given a black-box model with F1 score F_1^{bb} trained on the same features, the irreducible complexity gap at depth D is $G_D = F_1^{\text{bb}} - C_D$.*

$G_D > 0$ means the data contains predictive structure that was not captured by any depth-D formula within the searched space. Because the search is exhaustive within each phase template but does not cover all possible depth-D formulas, G_D is an upper bound on the true gap—a formula structure not in our template set could in principle achieve higher F1. However, if C_D is stable across multiple operation sets and phase structures, this suggests (but does not prove) that the gap is intrinsic to the depth constraint rather than a limitation of the search.

Proposition 1. *C_D is non-decreasing in D and bounded above by F_1^{Bayes}. If $C_{D+1} = C_D$ for a sufficiently rich operation set, the ceiling has converged.*

3 Benchmark Validation

To validate the method beyond the three-body case study, we run the phased search on four standard classification datasets and compare the formula ceiling to gradient-boosted ensembles

(Table 1).

Table 1: Formula ceiling across domains. "Gap" $= F_1^{\mathrm{GB}} - C$. Negative gap means the formula outperforms the ensemble.

Dataset (d)	Ceiling	GB F1	Gap	Formula
Circles (2)	**0.996**	0.989	−0.007	`hypot(x1,x2)`
Breast Ca. (30)	0.948	0.962	+0.014	`f7 + f22`
Ionosphere (34)	0.902	0.934	+0.032	`log(f6 min f5)`
Moons (2)	0.833	0.969	+0.136	`x2`
Three-body (23)	0.480	0.860	+0.380	Eq. **??**

On **Circles**, the method discovers the exact analytic decision boundary $\sqrt{x_1^2 + x_2^2} > \tau$ and *outperforms* gradient boosting ($F_1 = 0.996$ vs. 0.989). On **Breast Cancer** and **Ionosphere**, simple two-feature formulas achieve 95–98% of ensemble performance. On **Moons**, the nonlinear boundary resists depth-3 formulas (gap = 0.136). On the **Three-body** problem, the gap is largest (0.38), reflecting genuine nonlinear complexity.

The ceiling concept is validated: it correctly identifies problems that admit simple formulas (Circles: gap < 0), problems where formulas are close (Breast Cancer: gap = 1.4%), and problems that fundamentally require complex models (Three-body: gap = 38%).

4 Case Study: Three-Body Periodic Orbit Prediction

4.1 Setup

We use two datasets. The *formula search dataset* consists of 15,000 random configurations of equal-mass bodies ($m_1 = m_2 = m_3 = 1$) used for the phased search. The *ensemble comparison dataset* consists of 51,000 configurations spanning three mass ratios (equal, binary+light, hierarchical) used to train and cross-validate the gradient-boosted baseline. Both datasets use the same 23 Hessian-derived features and the same orbit integration protocol.

Features include: numerical matrix rank (threshold $\epsilon = 10^{-6}$), potential block eigenvalues and their ratios, frequency cluster counts, spectral entropy, Hamiltonian energy, body separation ratios, and derived quantities. Ground truth (periodic vs. non-periodic) is obtained by symplectic leapfrog integration ($\Delta t = 0.005$, 8×10^4 steps, $T = 400$ natural time units) on an NVIDIA GV100 GPU, with periodicity defined as phase-space return to within 5% of initial extent after the first 10% of integration time. Batch size is maximized automatically using the batch-probe library [5].

4.2 Search Results

Table 2 summarizes the results across all four phases.

Table 2: Formula search results by phase (15,000 configs, 23 features). All phases search formulas of depth ≤ 3.

Phase (depth)	Best formula	F1	Tested
1: single+unary (1–2)	`n_clusters`	0.35	138
2: pairwise (2)	`n_clust +` $\lambda_{\min}$	0.48	3,542
3: unary of top-30 (3)	`hypot(n_cl,` $\lambda_{\min}$`)`	0.48	150
4: triple of top-8 (3)	(none above 0.30)	—	224
Best (depth ≤ 3)		**0.48**	**4,054**

The best F1 is reached at Phase 2 and does not improve in Phases 3–4. All 20 top formulas are algebraically equivalent to linear combinations of n_{clusters} and $\lambda_{\min}$. The Euclidean norm variant

$$\hat{y} = \mathbf{1}\left[\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2} > \tau\right] \tag{1}$$

is representative of the equivalence class.

4.3 Irreducible Complexity Gap

Table 3: Formula ceiling vs. black-box models. The ceiling is measured on the 15K search dataset; the ensemble is cross-validated on the 51K comparison dataset (3 mass ratios, same features).

Model	Dataset	F1	Gap
Random (base rate)	15K	∼0.07	—
$r_2/r_1 > 10$	15K	0.29	—
Best formula (Eq. 1)	15K	0.48	0.00
Decision tree (depth 5)	51K (CV)	0.58	−0.10
Gradient boosting	51K (CV)	0.86	−0.38

The observed gap is $G = 0.86 - 0.48 = 0.38$. The decision tree (F1 = 0.58) captures some of this gap via piecewise-constant splits; the remaining $0.86 - 0.58 = 0.28$ requires the nonlinear interactions of an ensemble. Within the formula structures searched, no expression of depth ≤ 3 achieves more than 56% of the ensemble's predictive power. This gap may narrow with deeper search or richer operation sets, but its stability across our tested operations suggests it reflects genuine complexity in the data.

4.4 Physical Interpretation

The discovered formula combines two independent necessary conditions:

- $n_{\text{clusters}} \geq 2$: the eigenvalue frequency spectrum separates into fast (inner binary) and slow (outer body) groups, indicating timescale separation.
- $\lambda_{\min} > 0$: the smallest Hessian eigenvalue is nonzero, indicating the system is gravitationally bound (not escaping).

The Euclidean norm indicates these contribute independently—there is no synergy or interaction between them. Periodicity requires *both* separated timescales *and* sufficient binding.

4.5 Comparison to PySR

We compare directly against PySR [4] on four AI Feynman regression problems (Table 4).

Table 4: Regression: our method vs. PySR (R^2, 3,000 samples).

Problem	Ours	PySR	PySR time	PySR formula
Gravity	0.99	**1.00**	21s	`m1*m2/r^2` (exact)
Coulomb	0.99	**1.00**	5s	`q1*q2/r^2` (exact)
$x_1^2+x_2^2$	0.50	**1.00**	6s	`sq(x1)+sq(x2)` (exact)
Thin lens	0.93	0.96	7s	approximate

PySR recovers the exact formula on 3/4 problems by exploring deeper trees (depth 10+). Our method matches PySR on Gravity and Coulomb (where depth 2 suffices: $m_1 m_2/r^2$) but misses $x_1^2 + x_2^2$ (which requires two additive squared terms—a structure outside our Phase 2 template). The key distinction: PySR produces a single best formula; our method produces the ceiling + gap diagnostic.

4.6 Convergence and Ablation

Table 5 shows the ceiling at each search phase.

Table 5: Ceiling convergence: F1 at each phase.

Dataset	Phase 1	Phase 2	Phase 3
Circles	0.730	**0.996**	0.996
Moons	**0.833**	0.833	0.833
Breast Cancer	0.926	**0.934**	0.934

The ceiling converges at Phase 2 for all datasets. Phase 3 (unary transforms) never improves the best score, confirming that the ceiling is a stable property rather than a search artifact.

Ablation. On Circles, removing the `hypot` operation drops the ceiling from 0.996 to 0.796. All other operations are redundant—`hypot` alone captures the circular decision boundary. This demonstrates that the ablation identifies which operations are structurally necessary.

4.7 Runtime Scaling

Table 6: Runtime vs. feature count (Phase 1+2, single CPU core).

d	Formulas	Time (s)	Best F1
2	16	0.01	0.96
5	145	0.09	0.98
10	640	0.38	0.96
15	1,485	0.91	0.97
20	2,680	1.66	0.95

Runtime scales as $O(d^2)$, reaching 1.7 seconds at $d = 20$. GPU vectorization (not used here) would further reduce these times. At $d = 50$, the estimated Phase 2 time is $\sim$20 seconds on CPU.

5 Discussion

5.1 When to Use This Method

The phased exhaustive approach is most valuable when: (i) the feature count is moderate ($d < 50$, so d^2 is tractable); (ii) the practitioner needs the best formula within a bounded depth, not a stochastic approximation; (iii) the formula ceiling and irreducible complexity gap are of interest (e.g., to decide whether a problem admits a simple law or requires a complex model).

For large feature spaces ($d > 100$) or deep formula trees (depth > 5), genetic methods (PySR) or neural symbolic regression (AI Feynman) are more appropriate. Our method's advantage is the ceiling diagnostic: PySR finds better formulas on deep problems, but does not quantify when further search is futile.

5.2 The Ceiling as a Scientific Diagnostic

The formula ceiling is not just a performance metric—it is a statement about the data. A converged ceiling means the relationship between features and labels has a *fundamental* nonlinearity that no short formula can capture. In our three-body case study, the ceiling of 0.48 at depth 4 with the gap of 0.38 to gradient boosting tells us that three-body periodic orbit prediction is genuinely complex: it depends on nonlinear feature interactions that the underlying physics does not reduce to a compact law at the level of Hessian features.

This diagnostic is novel. Existing symbolic regression methods report the best formula found but do not quantify *how much better* a more complex model could do. The ceiling provides a principled stopping criterion: if $C_{D+1} = C_D$ and $G_D > 0$, stop searching for formulas and use an ensemble.

5.3 Limitations

The search is exhaustive only within each phase's structure. Phase 2 (pairwise) is complete; Phase 3 applies unary transforms only to top-K pairwise results, not all; Phase 4 tests only pre-specified triple patterns. A formula with structure not matching any phase template (e.g., $\log(x_1 + x_2) \cdot x_3/x_4$) will be missed.

The method does not discover features—it evaluates pre-computed ones. The quality of the discovered formula depends entirely on the feature engineering.

The F1 metric weights precision and recall equally. Domain-specific objectives (e.g., prioritizing recall for safety applications) could change the optimal formula.

6 Conclusion

We have introduced a phased exhaustive search framework for symbolic mechanism discovery that provides two novel contributions: guaranteed-best formulas at each complexity level, and the formula ceiling as a diagnostic for irreducible complexity. Applied to the three-body problem, the method discovers that periodicity depends on two Hessian features (frequency cluster count and minimum eigenvalue) with a hard ceiling at F1 ≈ 0.48, while ensemble methods reach 0.86. The 38% irreducible complexity gap quantifies the fundamental nonlinearity of three-body dynamics at the level of Hessian features.

The method is domain-general, requiring only a feature matrix and binary labels. It is practical for feature spaces up to $d \approx 50$ and formula depths up to 4, with GPU acceleration enabling evaluation of ∼6,000 formulas in under one minute.

References

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